Digital Signal Processing

Linear Transform
Block transform

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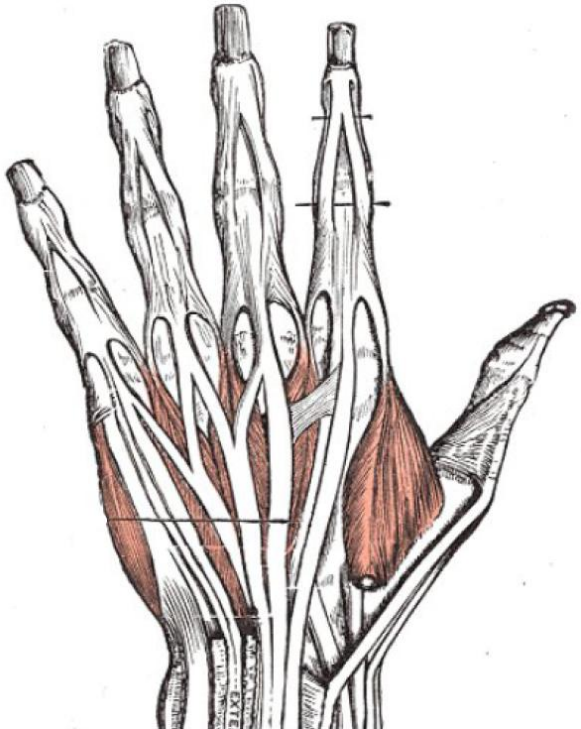
Transforms

We use transforms to see the same information from another point of view

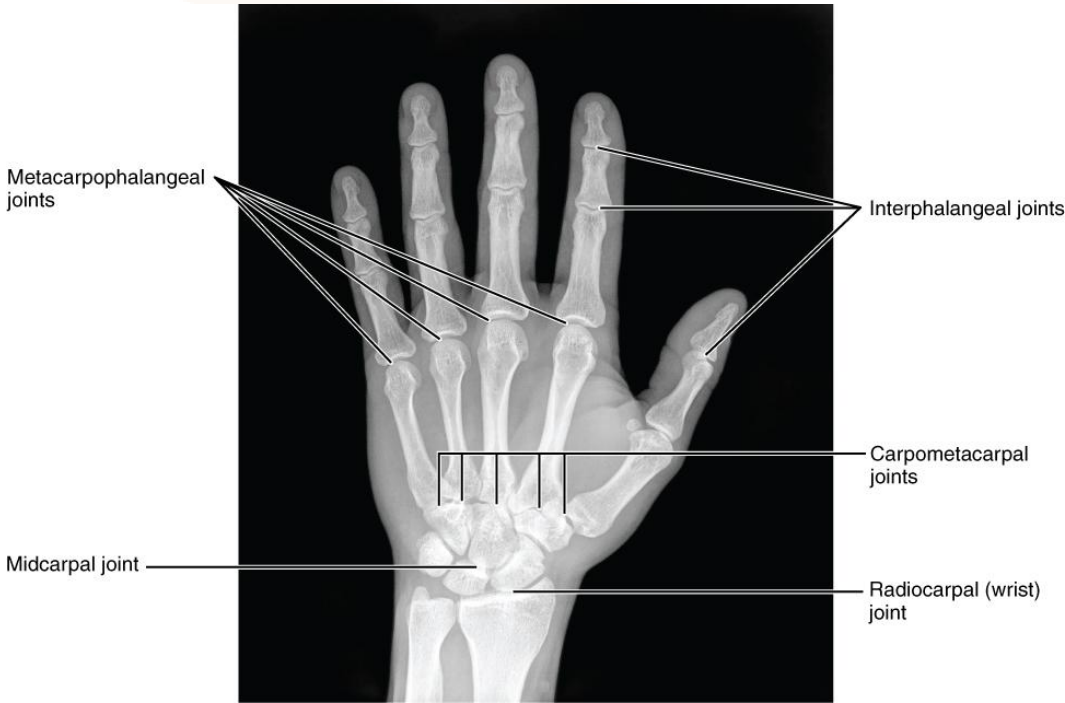
In telecommunications, we often decompose a time signal into its frequency components

Original signal H

D-transform of H



X-transform of H



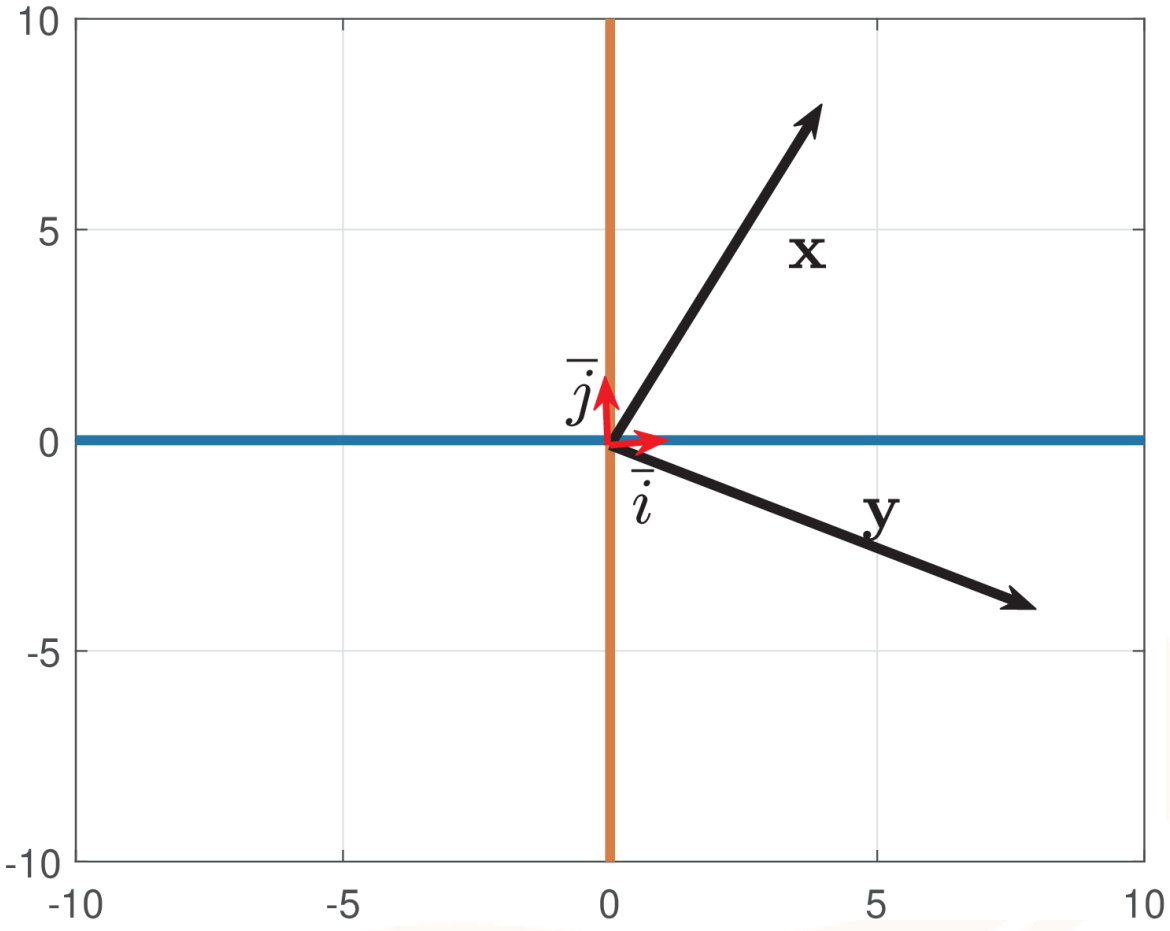
Any linear transformation can be represented as a matrix

$$y = Ax$$

In this case, the elements of y are interpreted as the *transform coefficients*, while x is interpreted as the original vector.

Example:
Clockwise rotation of θ rad:

$$\mathbf{A} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

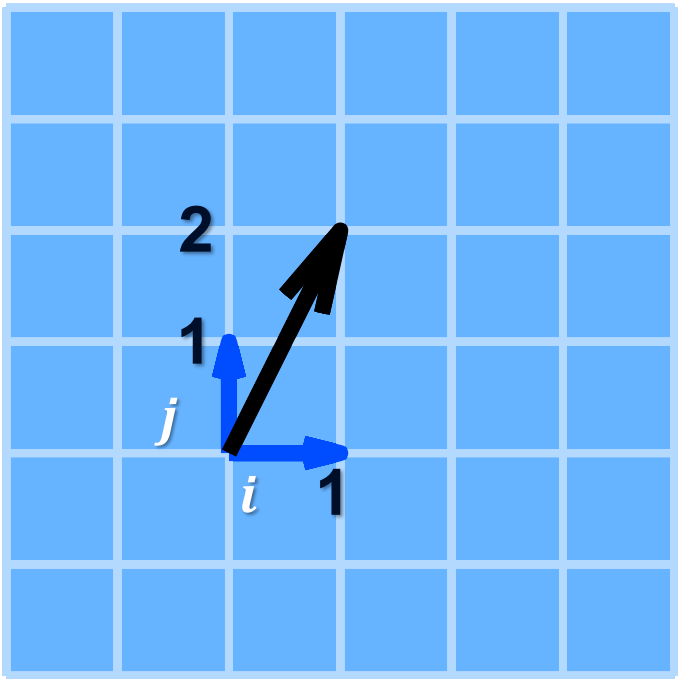


$x = [4, 8]$ was rotated by $\theta = \pi/2$, obtaining $y = [8, -4]$

An intuition for linear transforms

$$\bar{x} = 1\bar{\imath} + 2\bar{\jmath}$$

Standard Space (i, j)



Obtain transform coefficients

$$A = \frac{1}{3}\begin{bmatrix} 1 & 4 \\ -4 & 8 \end{bmatrix}$$

A

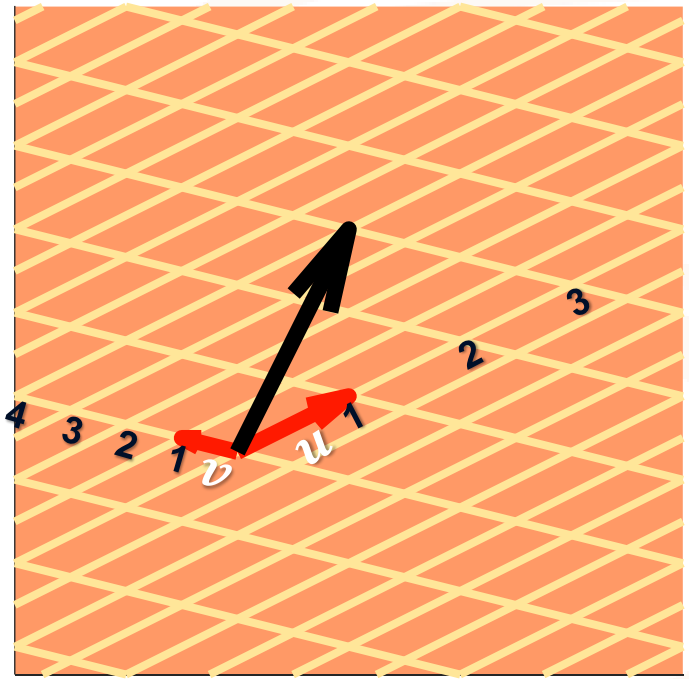
A^{-1}

Obtain original coefficients

$$A^{-1} = \frac{1}{8}\begin{bmatrix} 8 & -4 \\ 4 & 1 \end{bmatrix}$$

$$\bar{x} = 3\bar{u} + 4\bar{v}$$

Transformed Space (u, v)



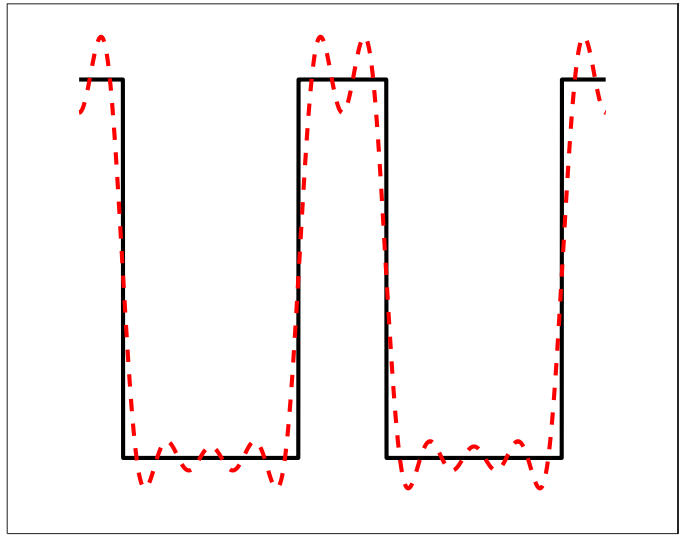
We transformed the basis $(\bar{\imath}, \bar{\jmath})$ into $(\bar{u}, \bar{v})$

An inverse transform may not always exist!

We can also see that (u,v) is not an orthonormal basis

Spoiler for Fourier Series

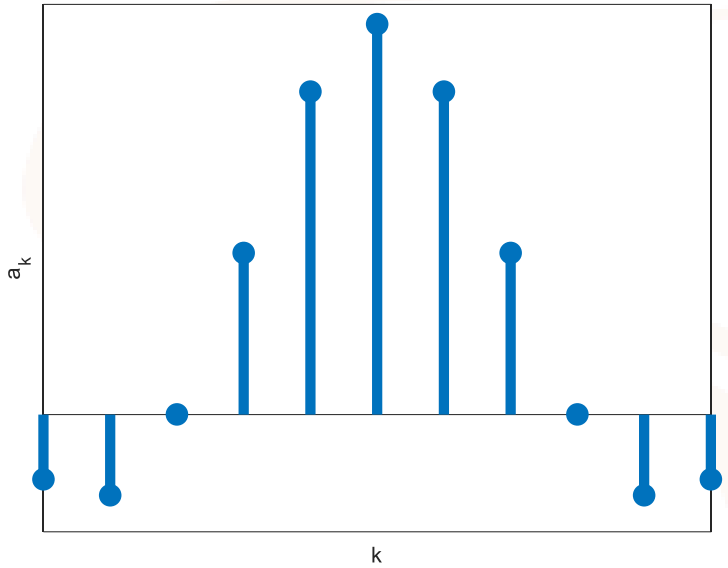
$$x(t) = x(t+T)$$



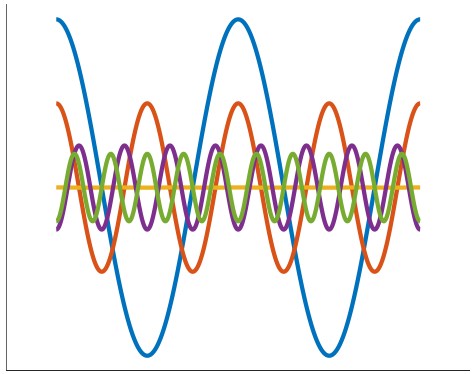
A

A^{-1}

$$x(t) = \sum_k a_k f_k(t+T)$$



A periodic time signal with period T can be represented by a sum of orthogonal periodic signals from a suitable basis



If the chosen basis signals are complex exponentials with common period, Fourier series allows us to compute the coefficients for each exponential

Inner product is the generalization of dot product

Most transforms are designed such that "transforming" corresponds to the calculation of inner products

The inner product $\langle \cdot,\cdot \rangle$ is a generalization of the dot product between real vectors, to complex signals and vectors

$$\langle \mathbf{x}, \mathbf{y} \rangle \triangleq \| \mathbf{x} \| \, \| \mathbf{y} \| \cos(\theta)$$

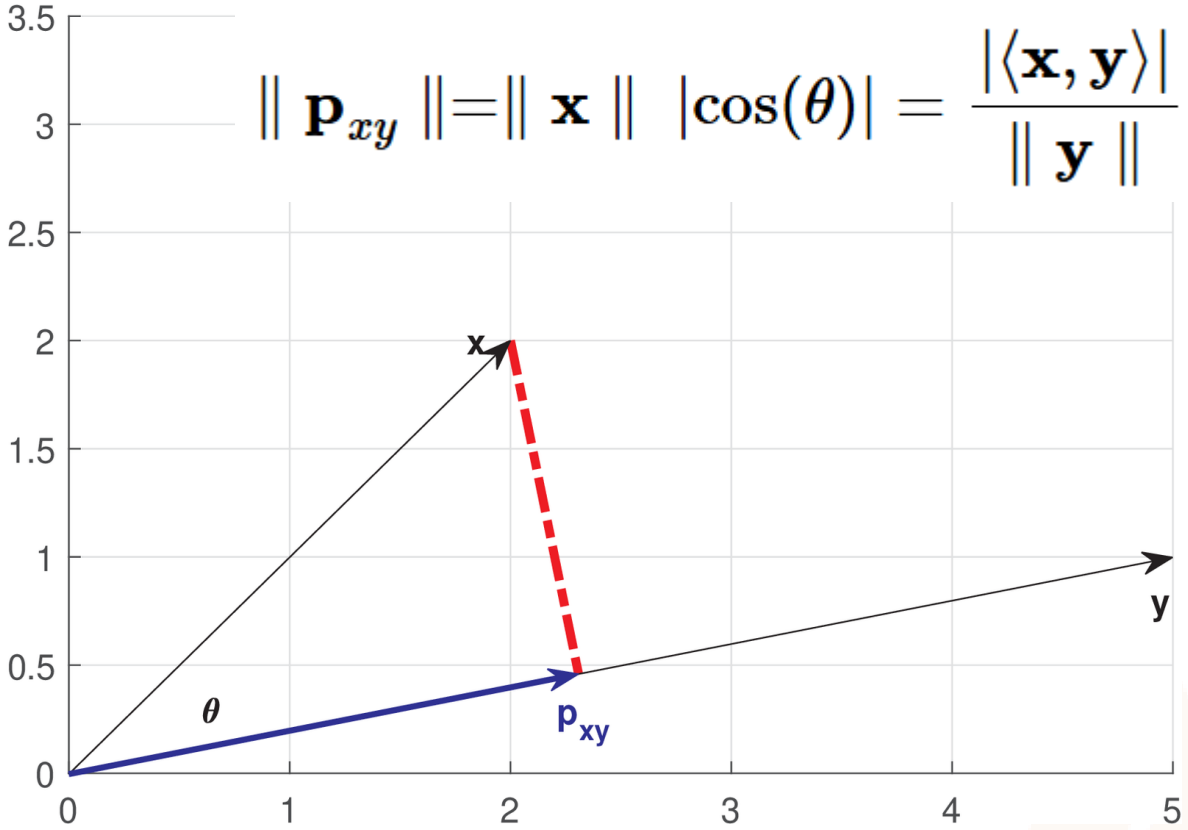
$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{N} x_i \, y_i = x_1 y_1 + x_2 y_2 + \cdots + x_N y_N$$

$$\langle x[n], y[n] \rangle \triangleq \sum_{n=-\infty}^{\infty} x[n] y^*[n]$$

$$\langle x(t), y(t) \rangle = \int_{-\infty}^{\infty} x(t) y^*(t) \mathrm{d}t$$

$$\langle x(t), y(t) \rangle = \int_{\langle T \rangle} x(t) y^*(t) \mathrm{d}t$$

$$\| \mathbf{p}_{xy} \| = \| \mathbf{x} \| \; |\cos(\theta)| = \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|}{\| \mathbf{y} \|}$$



If vector y has unit magnitude, the inner product $\langle x, y \rangle$ equals to the coefficient of x in the direction of y

Orthogonal is the generalization of perpendicular

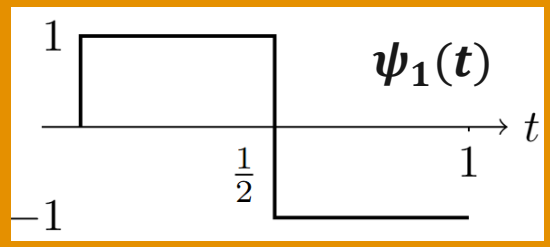
An orthogonal basis is a set of functions (or vectors) where every inner product between different elements is $\langle k, q \rangle = 0, k \neq q$

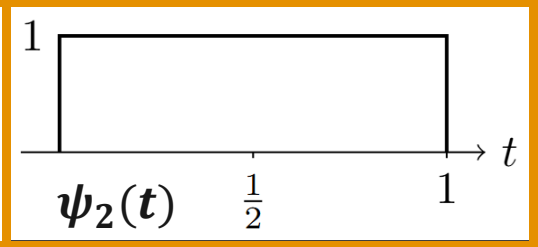
The set $\mathrm{F} = \{1, cos(k\omega_0 t), sin(k\omega_0 t): k \in \mathbb{N}\}$ is orthogonal of period T_0

Moreover, if $\langle k, k \rangle = 1$ for all k, the set is also called orthonormal

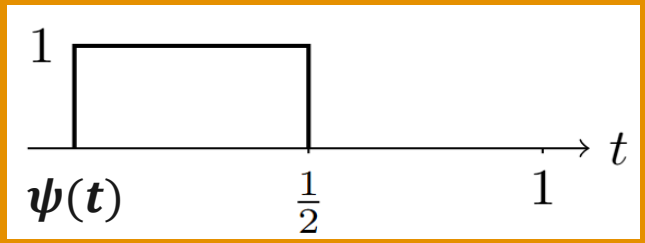
The set $\mathrm{A} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is orthonormal

The functions below can be used as an orthonormal set





The following $\psi(t)$ can be represented in terms of the elements of the set above

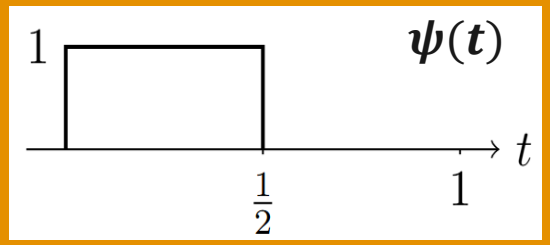


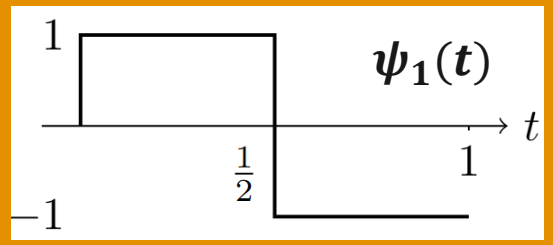
How can we calculate the coefficients?

Computing transform coefficients using inner product

Remind:

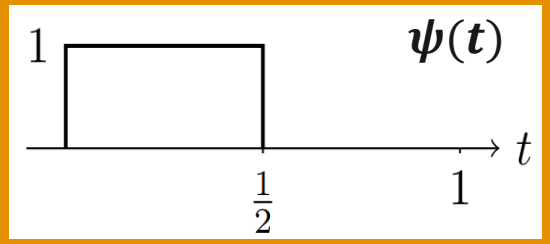
If vector y has unit magnitude, the inner product $\langle x, y \rangle$ equals to the coefficient of x in the direction of y

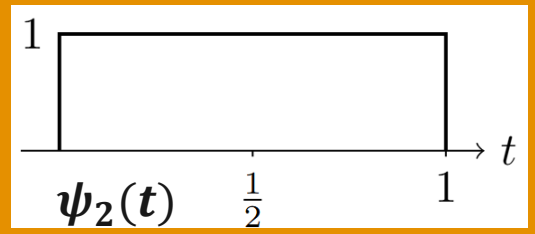
$\langle$ 

, 

$\rangle$

$$a_1 = \int_0^1 \psi(t)\psi_1^*(t)dt = 0.5$$

$\langle$ 

, 

$\rangle$

$$a_2 = \int_0^1 \psi(t)\psi_2^*(t)dt = 0.5$$

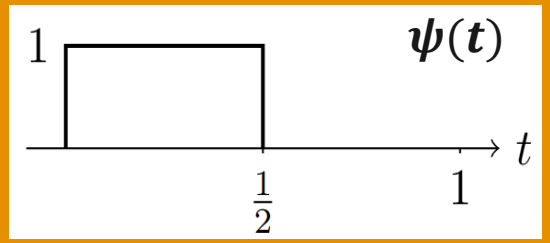
Since the set is orthonormal

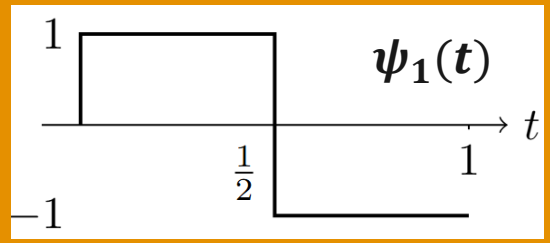
$$\psi(t) = 0.5\psi_1(t) + 0.5\psi_2(t)$$

More complicated solution for non orthogonal set

Non orthogonal set as a basis

Let $\psi_3(t) = 0.6\psi_1(t) + 0.8\psi_2(t)$ with unit energy

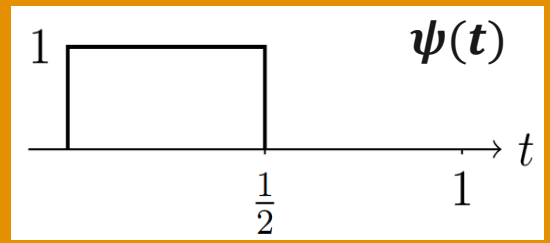
$\langle$ 

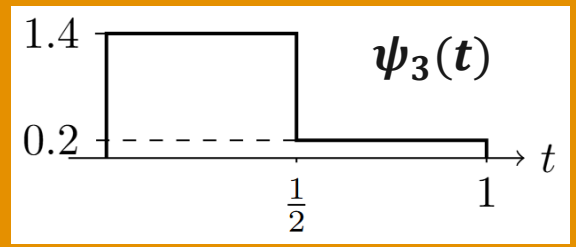
, 

$\rangle$

Our new set is $\{\psi_1(t), \psi_3(t)\}$

$$a_1' = \int_0^1 \psi(t)\psi_1^*(t)dt = 0.5$$

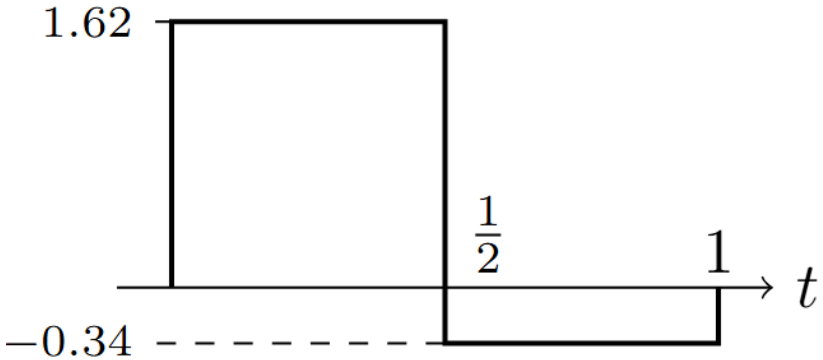
$\langle$ 

, 

$\rangle$

$$a_3' = \int_0^1 \psi(t)\psi_3^*(t)dt = 0.7$$

$$\psi'(t) = 0.5\psi_1(t) + 0.8\psi_3(t)$$



We were not able to recover $\psi(t)$!

Solving for non orthogonal basis

$$\Psi = [\psi_1, \psi_3] = \begin{bmatrix} 1 & 0.6 \\ 0 & 0.8 \end{bmatrix}$$

$$a_1' = \langle \psi, \psi_1 \rangle = \langle a_1\psi_1 + a_3\psi_3, \psi_1 \rangle$$
$$= \boxed{a_1\langle \psi_1, \psi_1 \rangle + a_3\langle \psi_3, \psi_1 \rangle}$$

Gram matrix
$G = \Psi^H \Psi$

$$G = \begin{bmatrix} \langle \psi_1, \psi_1 \rangle & \langle \psi_3, \psi_1 \rangle \\ \langle \psi_1, \psi_3 \rangle & \langle \psi_3, \psi_3 \rangle \end{bmatrix}$$

The formulation is
$a' = Ga$

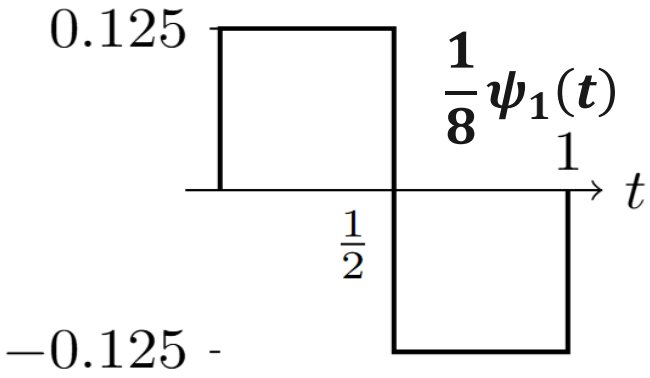
The solution is
$G^{-1}a' = a$

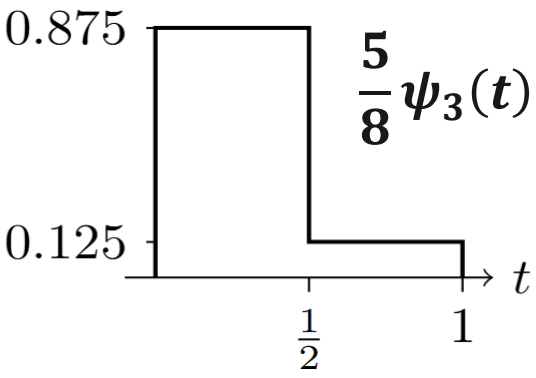
$$G = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 1 \end{bmatrix}$$

$$G^{-1} = \frac{1}{16}\begin{bmatrix} 25 & -15 \\ -15 & 25 \end{bmatrix}$$

Coefficients are $[a_1, a_3] = \left[\frac{1}{8}, \frac{5}{8}\right]$
$\psi(t) = \frac{1}{8}\psi_1(t) + \frac{5}{8}\psi_3(t)$ (Verify!)

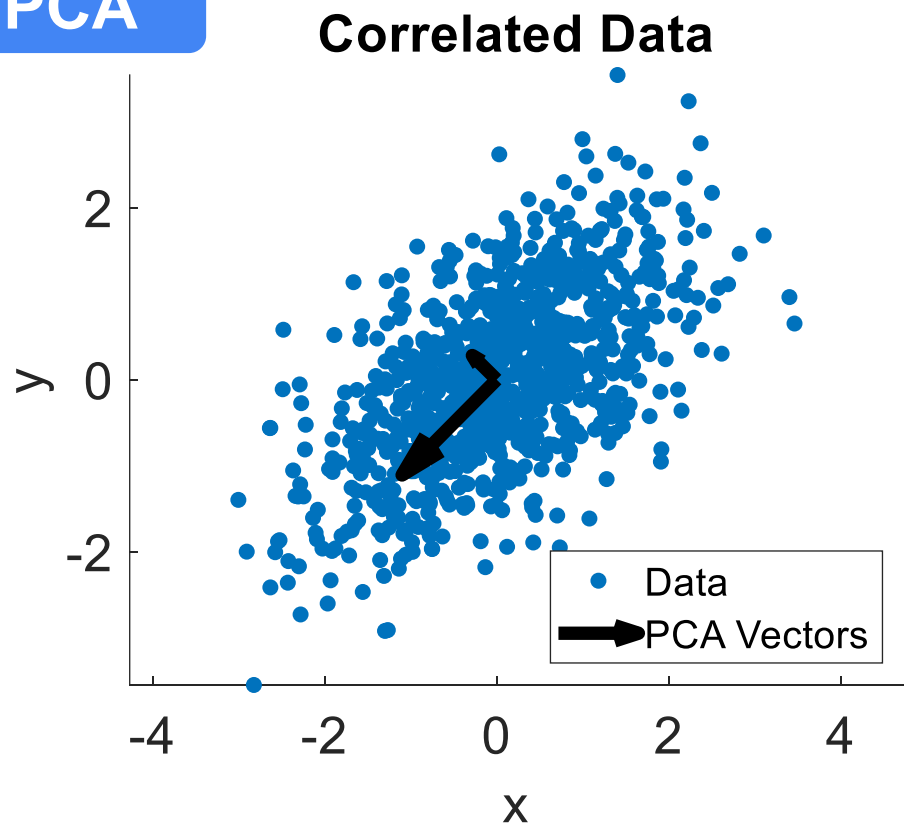
The task:
We don't know the true a_1 and a_3
But we know $a_1' = \langle \psi, \psi_1 \rangle = 0.5$
and $a_3' = \langle \psi, \psi_3 \rangle = 0.7$





Some orthogonalization procedures

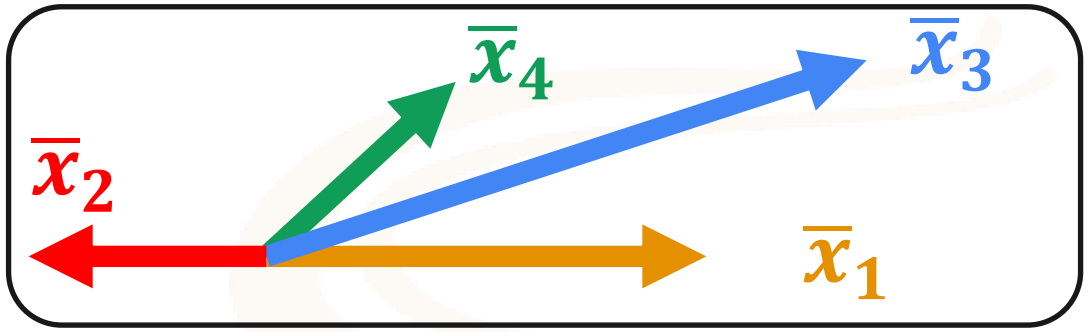
PCA



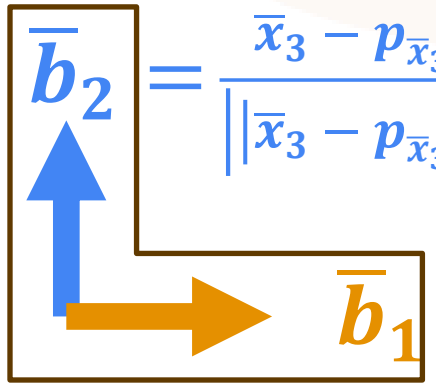
PCA finds orthonormal vectors and singular values from covariance matrix

Gram-Schmidt

$X =$



$B =$



$$\bar{b}_2 = \frac{\bar{x}_3 - p_{\bar{x}_3\bar{b}_1}}{\left\|\bar{x}_3 - p_{\bar{x}_3\bar{b}_1}\right\|}$$

$$\bar{b}_1 = \bar{x}_1 / \|\bar{x}_1\|$$

Subtract all projections and make it unit norm

Grand-Schmidt finds orthonormal basis B *iteratively* from the set of vectors X

Unitary matrices: Energy conservation

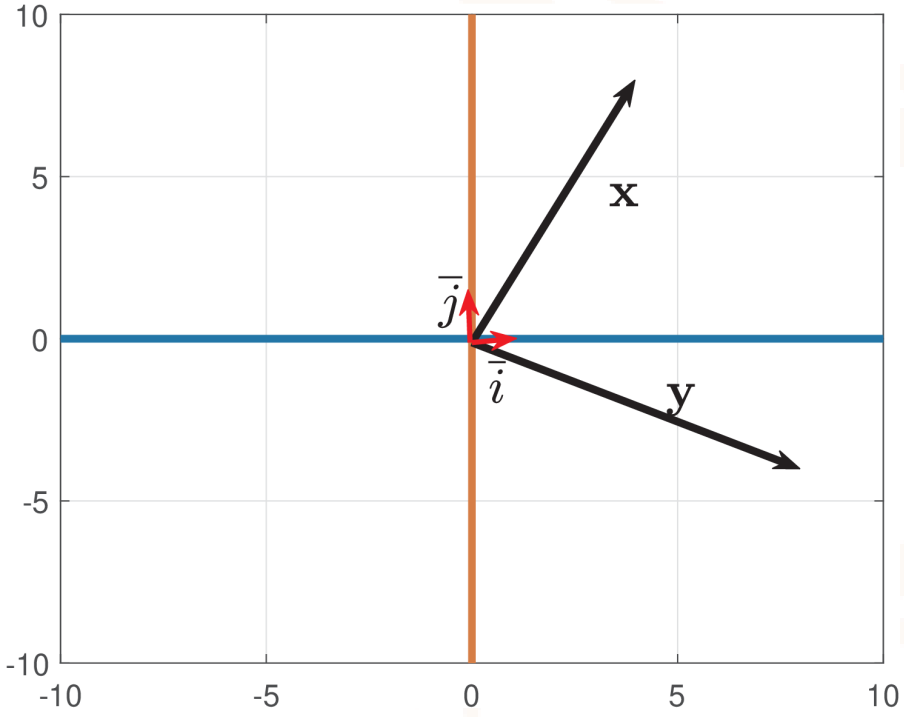
If the columns of a matrix B are orthonormal, then the matrix is called unitary

Unitary matrices are widely used in transforms because their inverse is simply
$B^{-1} = B^H = (B^*)^T$

If a transform A is unitary, then
$||X|| = ||Ax|| = ||x||$
Parseval theorem
$||X||^2 = ||x||^2$

The rotation matrix is unitary

$$\mathbf{A} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$



Early example:
$x = [4, 8]$
$y = [8, -4]$
$||x||^2 = ||y||^2 = 80$ J

Unitary transforms: Forward and Inverse matrices

The forward matrices contain the complex conjugate basis functions in their rows

$$y = Ax = \begin{bmatrix} a_0^*[0] & a_0^*[1] & a_0^*[2] \\ a_1^*[0] & a_1^*[1] & a_1^*[2] \\ a_2^*[0] & a_2^*[1] & a_2^*[2] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \end{bmatrix} = \begin{bmatrix} \langle x, a_0 \rangle \\ \langle x, a_1 \rangle \\ \langle x, a_2 \rangle \end{bmatrix}$$

Interpretation: coefficient 0 is the inner product of x with basis function $a_0[n]$

The inverse matrices contain the basis functions in their columns

$$x = A^{-1}y = \begin{bmatrix} a_0[0] & a_1[0] & a_2[0] \\ a_0[1] & a_1[1] & a_2[1] \\ a_0[2] & a_1[2] & a_2[2] \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x[0] \\ x[1] \\ x[2] \end{bmatrix}$$

Interpretation: to find the original $x[0]$, combine all the basis elements $\mathrm{b_n}[0]$ using the transform coefficients

Orthogonal matrices: easy inversion even if not unitary

Suppose an orthogonal matrix has basis vectors with magnitudes 2, 4 and 5 as the column vectors

$$A = \begin{bmatrix} \sqrt{2} & -2\sqrt{2} & 0 \\ \sqrt{2} & 2\sqrt{2} & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

We can extract those magnitudes with another matrix, obtaining a unitary matrix on the right side

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverting A means inverting the fractions in the first matrix, then performing $(.)^H$ on the second matrix

$$A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/5 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -\sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

If the original magnitudes were all equal to some value K, inversion is easier: a scalar instead of a matrix

$$A = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} A^H$$

DCT

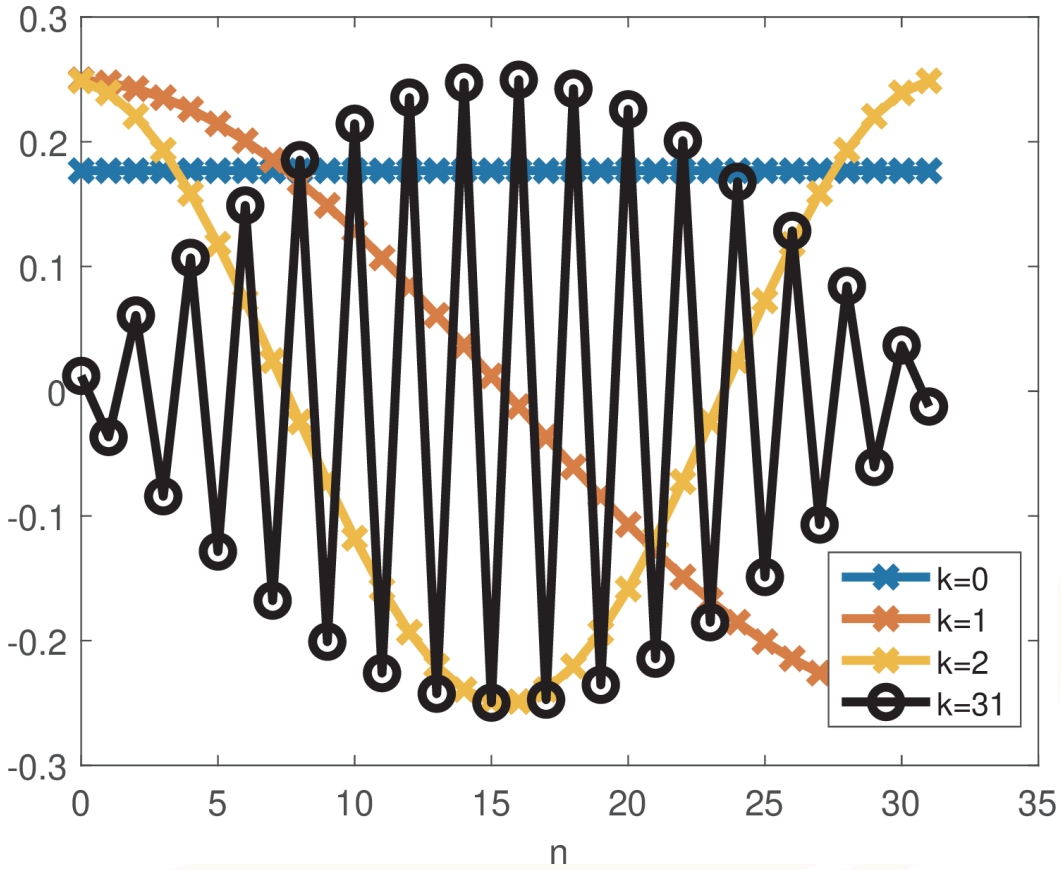
The Discrete Cosine Transform is a unitary matrix transform

$N = 4$:

$$\mathbf{A} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1.307 & 0.541 & -0.541 & -1.307 \\ 1 & -1 & -1 & 1 \\ 0.541 & -1.307 & 1.307 & -0.541 \end{bmatrix}$$

$$\mathbf{A}^{-1} = \mathbf{A}^T = \frac{1}{2}\begin{bmatrix} 1 & 1.307 & 1 & 0.541 \\ 1 & 0.541 & -1 & -1.307 \\ 1 & -0.541 & -1 & 1.307 \\ 1 & -1.307 & 1 & -0.541 \end{bmatrix}$$

$N = 32$:



In general: $a_{n,k} = w_k \cos\left(\frac{\pi(2n+1)k}{2N}\right)$

$w_k = 1/\sqrt{N}$ for $k = 0$,
$w_k = \sqrt{2/N}$ for $k \in \{1, .., N-1\}$

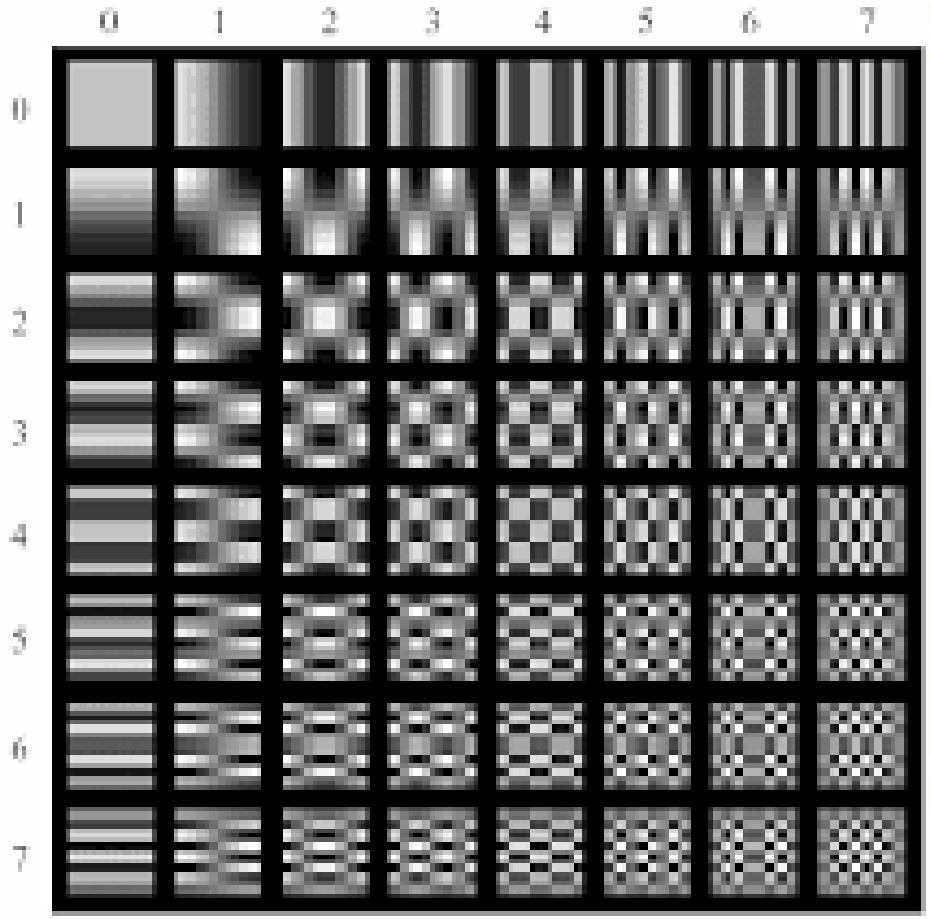
DCT: JPEG

In JPEG, the image channels Y, Cb and Cr (instead of RGB) are divided in blocks of 8x8

The 2D DCT is performed on every block, then information is compressed by quantization

High frequency coefficients are often quantized to zero, and there are techniques to exploit this redundancy

Basis functions of the 2D DCT:



DFT

The Discrete Fourier Transform is an orthogonal matrix transform

$$\frac{1}{N}e^{\frac{j2\pi nk}{N}} = \frac{1}{N}\left[\cos\left(\frac{2\pi nk}{N}\right) + j\sin\left(\frac{2\pi nk}{N}\right)\right]$$

The basis functions are complex exponentials

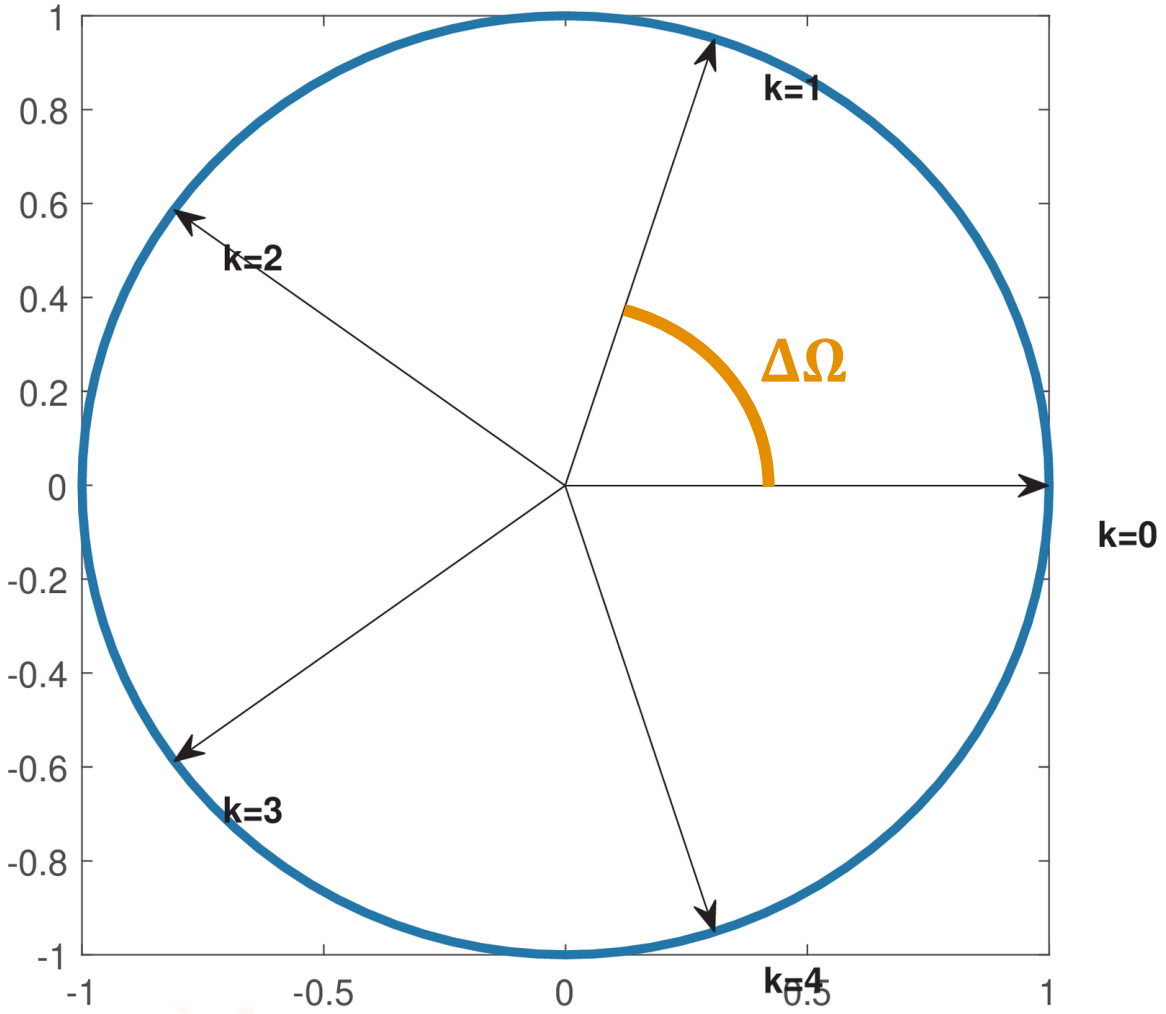
$$b[n] = \frac{1}{N}e^{j\Omega_k n} = \frac{1}{N}\left[\cos\left(\Omega_k n\right) + j\sin\left(\Omega_k n\right)\right]$$

For N=5, k=3, the basis function is

$$b_3[n] = \frac{1}{5}e^{j\left(\frac{2\pi 3}{5}\right)n},\ \Omega_k = \frac{2\pi 3}{5}$$

The angular resolution of the DFT is $\Delta\Omega = \frac{2\pi}{N}$

For $N = 5$: $\Delta\Omega = 2\pi/5$



DFT: Inverse and Direct transform

$$\mathbf{A}^{-1} = \frac{1}{N}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{j\pi/2} & e^{j\pi} & e^{j3\pi/2} \\ 1 & e^{j\pi} & e^{j2\pi} & e^{j3\pi} \\ 1 & e^{j3\pi/2} & e^{j3\pi} & e^{j9\pi/2} \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

The inverse matrix has the basis functions in its columns

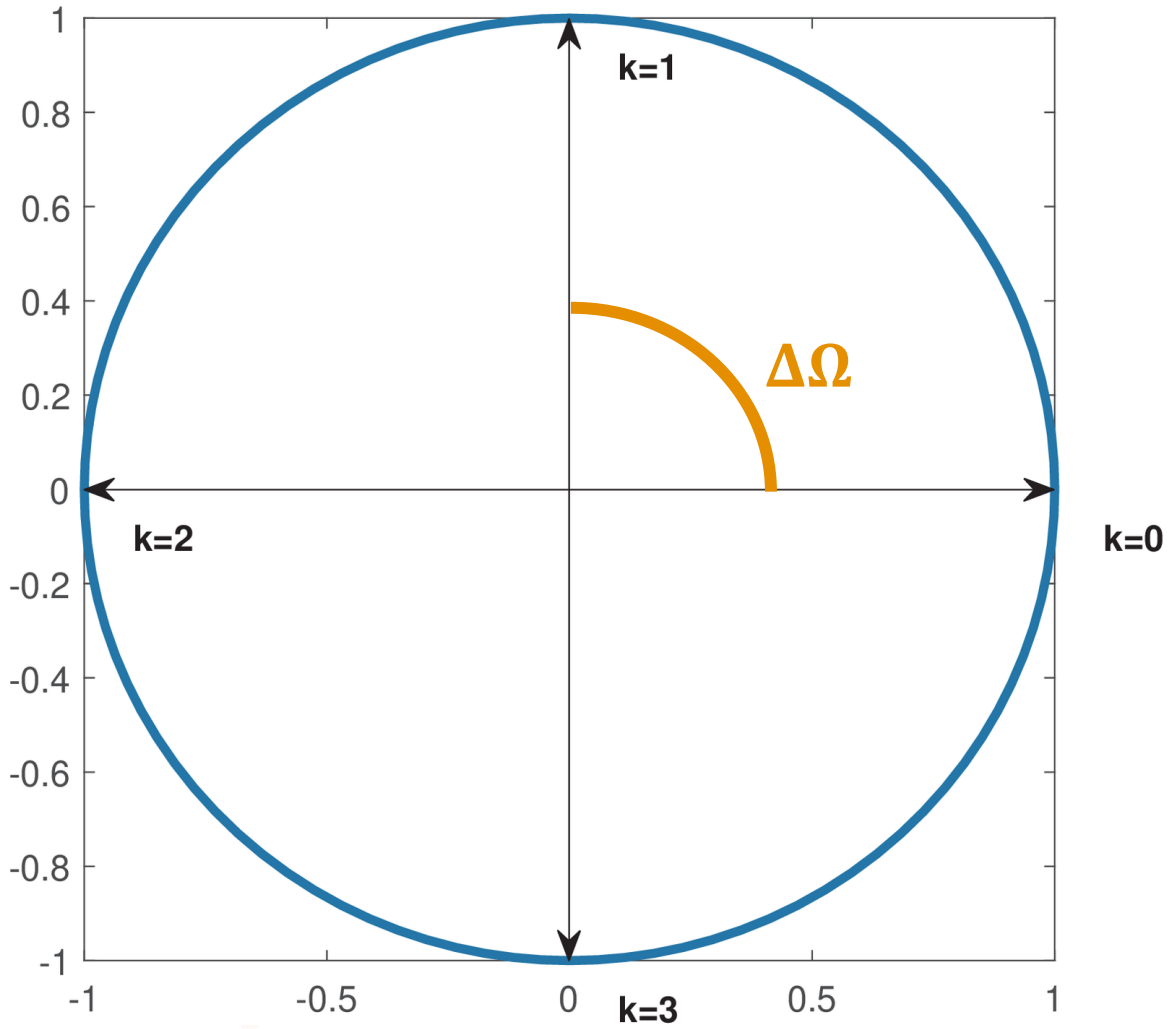
Direct coefficients
$H[k]$
$k \in \{0, \dots, N-1\}$

$$X[k] = \langle x[n], e^{\frac{j2\pi nk}{N}} \rangle$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi nk}{N}}$$

For $N = 4$:

$\Delta\Omega = \pi/2$



DFT coefficients in the unit circle

Positive Frequencies

Negative Frequencies

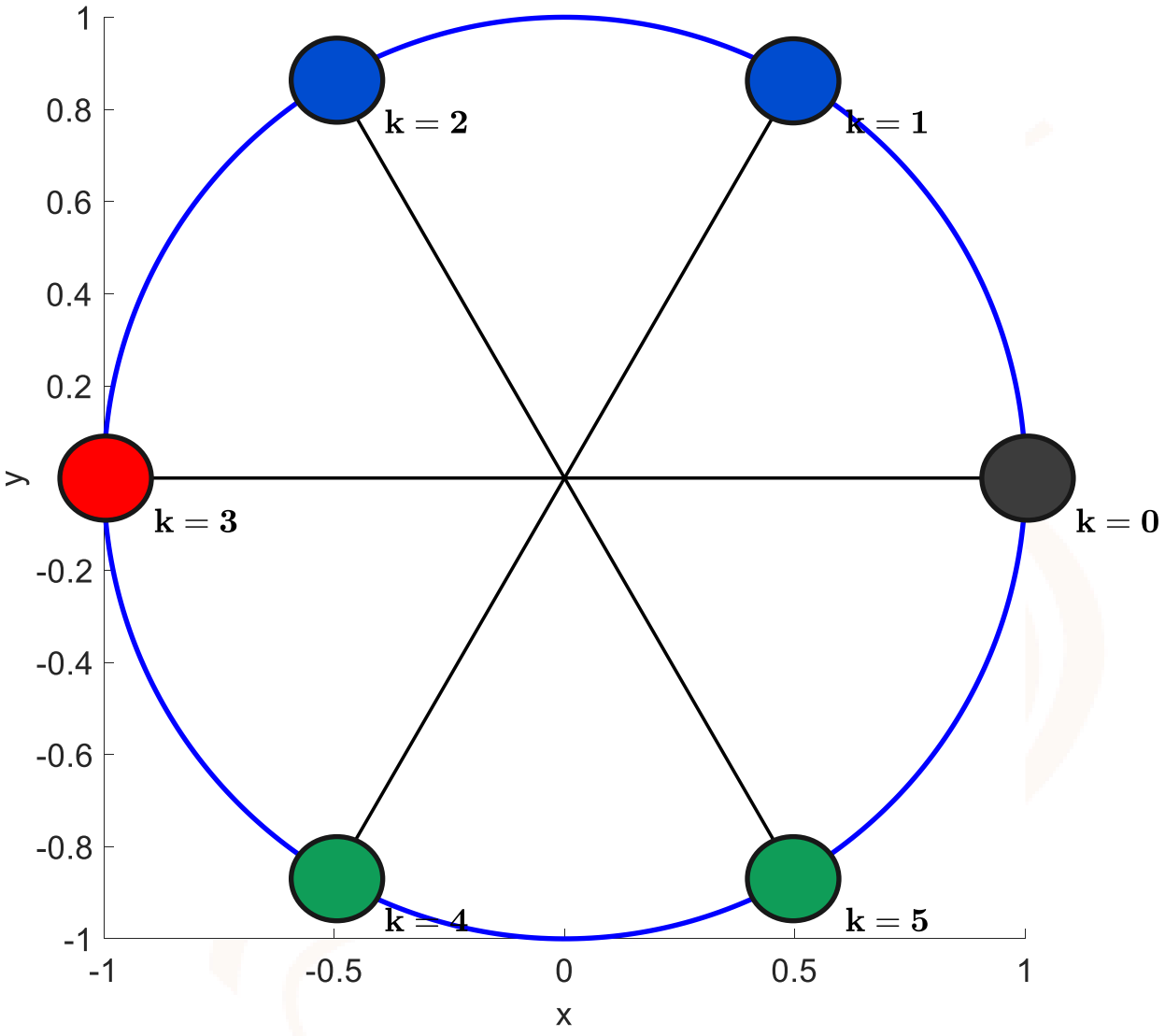
$k =$	0	1	2	3	4	5
$X[k] =$	0	1j	0.5	0.1	0.5	-1j
	DC			Nyquist		

Note 1: the Nyquist coefficient only exists if the number of elements is even

Note 2: when the original vector is real, the DFT coefficients have Hermitian symmetry

Use your knowledge to find the original vector x[n] from the DFT coefficients above

$\Delta\Omega = ?$



Quiz Time!

Linear transforms

Projections

Parseval Theorem

DFT: direct and inverse

Thank you!

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